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OSAM-Fundus: A training-free, one-shot segmentation framework for optic disc and cup in fundus images

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ABSTRACT

Accurate segmentation of the fundus optic disc and optic cup is essential for the early detection of glaucoma, a leading cause of blindness. Despite the advancements in deep learning-based segmentation models, the reliance on large amounts of annotated data and the weak generalization ability remain a challenge. The Segment Anything Model (SAM) has demonstrated robust zero-shot generalization in various natural image segmentation tasks. However, its dependency on manual interactive prompts and the domain gap between natural and medical images are obstacles to its application in medical image segmentation. This study introduces OSAM-Fundus, a novel training-free framework for optic disc and optic cup segmentation using one reference image and its annotation. First, We propose an automatic prompt generation method that incorporates medical prior knowledge and hard-negative sampling into patch-level bidirectional matching, enabling cross-image and cross-domain prompt generation without manual clinician intervention. Then, to overcome SAM's inherent challenge in segmenting fuzzy edges, we employ a feature re-matching strategy to refine the false-positive regions potentially produced by SAM, ensuring consistent semantic object outputs in the target image. Evaluation on four public datasets, both in-domain and out-of-domain, our method achieves state-of-the-art performance with average dice scores of 80.43%, which is 7.28% higher than that of SAM using prompts from ground truth. This demonstrates that OSAM-Fundus is a highly efficient and robust segmentation method for fundus images.

1. Introduction

Accurate segmentation of medical images is essential for computer-aided diagnosis of various diseases. In the context of fundus imaging, accurate segmentation can help diagnose a wide range of ocular diseases, including glaucoma, diabetic retinopathy, and age-related macular degeneration [1]. The typical symptoms of glaucoma, the second most common blinding disease worldwide, are often accompanied by morphological changes in the optic disc (OD) and optic cup (OC). Accurate measurement of these parameters is therefore essential for glaucoma diagnosis and treatment planning [2]. However, manual measurement by ophthalmologists is labor-intensive, time-consuming, and prone to errors, particularly in low contrast or vascular interference cases. Therefore, the development of an automated, reliable method for OD and OC segmentation to measure the parameters is crucial.

Automatic OD and OC segmentation methods are mainly categorized into traditional and deep learning methods. Traditional methods typically rely on the grayscale or color information of the image and the edge detection results, and gradually refine the contour position to obtain an accurate contour of the target object [3–5]. Although these methods have improved in segmentation effectiveness, remain highly dependent on specific features and thus perform poorly in pathological regions and low-contrast images. Recent deep learning-based methods have shown remarkable success by using features automatically learned from neural networks. Zilly et al. [6] used ensemble learning based on convolutional neural network (CNN) architecture to segment both OD and OC. Mohan et al. [7] replaced the pooling layer with atrous convolution to perform efficient feature extraction based on full-resolution residual networks. The fully convolutional network (FCN) [8] replaces the fully connected layer in CNN with a convolutional layer, which can adapt to input images of variable sizes and has therefore been widely used in image segmentation tasks. Shankaranarayana et al. [9] proposes a fully convolutional network that enhances optic disc and cup segmentation by utilizing estimated depth information alongside traditional color fundus images. FCN-based U-Net [10] is regarded as the most advanced and widely used model in medical image segmentation. Inspired by the success of U-Net, many variants have been used for OD and OC segmentation. M-Net [11], W-Net [12], and RMSDSC-Net [13] effectively capture image features at different scales by incorporating

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Table 1
Previous works and performance metrics obtained.

Author	Year	Method	Drishti-GS		RIM-ONE-r3		REFUGE/Test	
			OD	OC	OD	OC	OD	OC
Joshi et al. [3]	2011	Active Contour Model	85.00	80.00	87.00	72.00	–	–
Mary et al. [4]	2015	Active Contour Model	91.00	71.00	86.00	79.00	–	–
Haleem et al. [5]	2017	Adaptive Edge Smoothing	95.00	81.00	91.00	89.00	–	–
Zilly et al. [6]	2017	CNN	97.30	87.10	–	–	–	–
Mohan et al. [7]	2018	Residual Network	96.40	–	–	–	–	–
Fu et al. [11]	2018	Modified U-Net	–	–	–	–	96.09	88.46
Shankaranarayana et al. [9]	2019	FCN	96.30	84.80	97.00	87.60	–	–
Yuan et al. [12]	2021	Modified U-Net	–	–	–	–	96.44	90.03
Luo et al. [18]	2021	Adversarial Learning	97.50	89.80	95.10	86.60	95.10	86.60
Zhou et al. [13]	2022	Modified U-Net	97.80	91.90	–	–	96.50	91.00
Yi et al. [14]	2023	Transformer U-Net	97.61	91.95	–	–	96.93	90.82
Domian adaptation[a]								
Wang et al. [19]	2019	Adversarial Learning	96.50	85.80	86.50	78.70	96.02	88.26
Zhou et al. [20]	2023	Adversarial Learning	97.80	91.40	–	–	96.50	91.50

Domian generalization[b]

Author	Year	Method	REFUGE/Train		Drishti-GS		RIM-ONE-r3		REFUGE/Val	
			OD	OC	OD	OC	OD	OC	OD	OC
Wang et al. [21]	2020	Feature Embedding	91.98	86.66	95.59	83.59	89.37	80.00	93.32	87.04
Liu et al. [22]	2021	Meta-Learning	94.50	85.51	95.37	84.13	87.52	71.88	93.37	83.94
Lyu et al. [23]	2022	Reinforcement Learning	92.55	87.86	95.47	85.93	92.28	81.24	93.55	86.94

[a] The experimental setup was to train together with REFUGE as the source domain and the other domains as the target domains.
[b] The experimental setup followed a Leave-one-domain-out set, where one domain is left out as the test set, while the remaining domains are used as the training set.

multi-scale inputs into the U-Net model, which improves the detail accuracy of the segmentation. Then, W-Net [12] and RMSDSC-Net [13] improve the skip connection, transforming them into contextual semantic extraction modules. This enhancement reduces the semantic gap between deep and shallow features and facilitates feature fusion. In recent years, the proposal of Visual Transformer has expanded the application of Transformer in computer vision. Currently, researches are integrating U-Net with Transformer and using self-attention mechanism to enhance U-Net's ability to capture contextual relationships in fundus image segmentation tasks [14]. Despite these advancements, most methods assume that the training and test images have the same distribution. In reality, public retinal image datasets, such as Drishti-GS [15], RIM-ONE-r3 [16], and REFUGE [17], exhibit noticeable appearance differences due to variations in scanner type, image resolution, light source intensity, and parameter settings. The segmentation performance degrades rapidly when the model is applied to new datasets with unknown domains that differ in distribution from the training dataset, a phenomenon known as domain shift. Existing research attempts to address the domain shift problem through adversarial learning [18], domain adaptation [19,20], or domain generalization techniques [21, 22]. Table 1 lists the various methods discussed above for segmenting OD and OC, along with the datasets they were applied to and their corresponding results. However, these methods face the challenge due to the need for large amounts of labeled samples from the source domain or even the target domain and substantial computing resources.

The Segment Anything Model (SAM) [24] recently demonstrated impressive zero-shot segmentation performance, indicating its potential in open-world image perception [25–28]. However, its application to medical image segmentation faces challenges [29–32], primarily due to significant domain differences between medical and natural images. Specifically, SAM lacks the ability to extract high-level semantic features, requiring manual prompts for target region segmentation, which necessitates deep expertise in medical imaging. Moreover, directly applying SAM to medical image segmentation results in subpar performance, as the blurred boundaries of medical images frequently cause an increase in false-positive segmentation. To address the limitations of SAM due to its lack of semantic information, Grounded-SAM [33] combines various vision models [24,34–36] to enable the segmentation of arbitrary objects in images based on textual descriptions. However, this method is applicable to object categories that have been pre-trained in vision models and cannot be directly applied to fundus image segmentation. In addition, the method uses the output of one model as the input for another, leading to error accumulation within the model. Therefore, we have revisited the approach of providing semantic information for SAM and achieve personalized semantic segmentation for downstream tasks.

We propose OSAM-Fundus, a training-free framework for segmenting OD and OC in fundus images only using one reference image and its annotation. We have developed an automatic prompt generation method that requires without manual intervention. Firstly, the prompt points in the reference image are propagated to the target image based on patch-level bi-directional feature matching. Then, we utilize the consistency of feature expression within the same anatomical structure in medical images to refine the positive prompts by removing points with low similarity. Subsequently, we obtain negative prompt points that can be easily confused with positive samples through hard-negative sampling, enhancing SAM's understanding of non-target areas. After generating the prompts for the SAM, we further designed the feature re-matching module for refining the initial segmentation results of the SAM, and ultimately generating more accurate segmentation masks. Further, our framework can be simply and efficiently extended to K-shot, enabling the model to benefit from a multi-reference setting.

This paper presents the following key contributions:

(1) The development of OSAM-Fundus, a training-free framework for fundus image segmentation that automatically generates prompts for SAM in the target image using only one reference annotation pair, without requiring input from a clinical expert.

(2) The incorporation of medical prior knowledge and hard-negative sampling into bidirectional feature matching for automatic prompt segmentation across images and domains, along with a feature re-matching module for refining segmentation results.

(3) The attainment of competitive results, with an average dice coefficient that is 7.28% higher than the baseline model. The method also exhibits robust generalization capabilities, outperforming classic fully-supervised models on out-of-domain datasets.

The remainder of this paper is outlined as follows. Section 2 presents an overview of the related work. Section 3 introduces the details of the proposed OSAM-Fundus framework. Experimental results are shown in Section 4. Section 5 gives a discussion, and Section 6 draws the conclusion.

2. Related work

In this section, we briefly survey previous works that are most related to ours, including few-/one-shot segmentation, foundation computer vision models, and prompt engineering for SAM.

2.1. Few-/one-shot segmentation

Few-shot segmentation (FSS) refers to the approach of performing semantic segmentation tasks with a limited number of annotated samples. Specifically, utilizing only one annotated sample is termed one-shot segmentation (OSS). Shaban et al. [37] proposed the pioneering work in this field, known as OSLSM. This approach employs a dual-branch structure that utilizes the support branch to generate classifier weights for prediction in the query branch. Inspired by prototypical networks [38], utilizing prototype representations to guide mask prediction in query images has become a popular paradigm in few-shot segmentation literature. On this basis, FSS/OSS methods can be classified into two major categories, class-wise and pixel-wise approaches, distinguished by the interaction method between the support-query pairs. Class-wise approaches represent objects within the support sample as a prototype through masked average pooling [39], guiding mask prediction in the query sample based on the similarity between the support prototype and the feature embeddings of the query sample [40–42]. Although class-wise approaches have proven effective, they overlook the pixel-level pairwise relationships and the spatial structure of features between support and query samples. Pixel-wise methods compensate for the loss of spatial information by densely computing pixel-pixel similarities to establish support-query correlations. For example, Min et al. [43] introduced learnable 4D convolutions to analyze the correspondence between support and query images. Following this, Hong et al. [44] replaced the 4D convolution with a 4D swin transformer to further enhance this correspondence. In addition, some benchmark datasets, such as PASCAL5i [37], COCO-20i [45], and FSS1000 [46] provide data support for research in FSS/OSS.

In few-/one-shot medical image segmentation, most methods focus on expanding the training set through data augmentation [47]. While effective, they require retraining the model when new categories need to be segmented. Recently, some research has turned to designing network architectures that do not require retraining the model. SENet [48] was the first to use a two-branch architecture for FSS/OSS of medical images. MprNet [49] improved upon SENet by facilitating information exchange between the two branches using a cosine similarity-based fusion module. Additionally, significant progress has been made in few-/one-shot medical image segmentation methods based on metric learning. Yu et al. [50] proposed a location-sensitive local prototype network to utilize spatial prior for few/one-shot image segmentation. Tang et al. [51] utilized the correlation between foreground and background to enhance contextual features around object boundaries. However, unlike natural images, medical images lack large-scale public datasets for pre-training segmentation models, which limits the accuracy of few-/one-shot medical image segmentation. Different from existing OSS methods, we propose a novel training-free approach that leverages rapidly evolving large pre-trained foundational models.

2.2. Foundation computer vision models

With the rapid development of deep learning technology, most contemporary vision frameworks follow the pre-training and fine-tuning paradigm [52]. Pre-trained basic models perform well in adapting to various downstream scenarios owing to their robust generalization ability. Contrastive Language-Image Pretraining (CLIP) [53] learns image representations from scratch on over 400 million image-text pairs, showcasing impressive zero-shot image classification capabilities. DINOv2 [54] employs self-supervised learning technology to expand ViT's pre-trained model in terms of data and model size, aiming to generate universal visual features that simplify fine-tuning downstream tasks. Notably, SAM [24] was recently pre-trained on a dataset of over 11 million images, emerging as a general foundational model for natural image segmentation. Using an interactive and prompt-driven approach, SAM has demonstrated impressive zero-shot capabilities in segmenting different subjects in real-world environments.

2.3. Prompt engineering for SAM

The core strength of SAM is its interactive segmentation paradigm: the model segments objects based on prompts given by the user, such as points and bounding boxes. To reduce manual intervention, efforts are being made to automatically generate prompts for SAM.

Existing prompt engineering for SAM can be divided into three categories, namely, pre-training-based methods, priori information-based methods, and one-shot prompt-based methods. Pre-training-based methods provide point or box prompts for SAM in the inference stage by utilizing a pre-trained prompt generator. Wang et al. [55] propose a spatially aware prototype network, which involves training a small number of samples to obtain a coarse segmentation of the query image, followed by sampling points from it as prompts for SAM. Na et al. [56] train U-Net as a prompt generator and perform a binary classification task on the mask values, selecting a certain number of positive and negative points as prompts based on a predefined probability threshold. Cao et al. [57] pre-train the prompt generator as an object detection method to identify specific objects in images and generate box prompts. However, such methods still require retraining the model for each new task, resulting in a loss of zero-shot generalization capability. Priori information-based methods utilize the properties of downstream tasks to automatically generate prompts. Li et al. [58] utilize prior information to fuse the prompts of the region of interest key points and grid key points based on the distribution and shape of the material's microstructure. However, this approach relies heavily on pre-defined rules, which lack flexibility when faced with data diversity.

Recently, a novel training-free paradigm based on one-shot prompting has been attracting attention. Zhang et al. [59] utilize the reference image and the corresponding mask to obtain an overall confidence map of the target image. Based on these confidence scores, two points are selected as positive and negative positional priors, which are then encoded as prompt tokens and fed into the decoder of SAM for segmentation. Liu et al. [60] introduces a novel method for feature matching using generic features of pre-trained models and exploits this matching for efficient and effective prompt segmentation. Although one-shot prompting methods have the advantage of not requiring training, they face limitations when applied directly to medical image segmentation. This is primarily because most large-scale visual models are pre-trained on natural images, and the significant differences between natural and medical images limit effective knowledge transfer. Specifically, the complex structures and boundaries present in medical images demand more precise semantic understanding, whereas features learned from natural images are often inadequate to meet this requirement. Motivated by these observations, we propose an efficient automatic prompt generation method that combines the feature consistency of the same anatomical structures in medical images. Through patch-level feature matching, we accurately generate positive and negative prompt points for guiding SAM segmentation. In addition, to address potential false-positive segmentation results stemming from SAM processing fuzzy boundaries in medical images, we introduce a feature re-matching module to refine the preliminary segmentation mask.

3. Method

OSAM-Fundus is a training-free, one-shot framework designed for fundus image segmentation by integrating the universal feature extraction model DINOv2 and the promptable model SAM. Given a reference image and its semantic mask, OSAM-Fundus can segment objects with

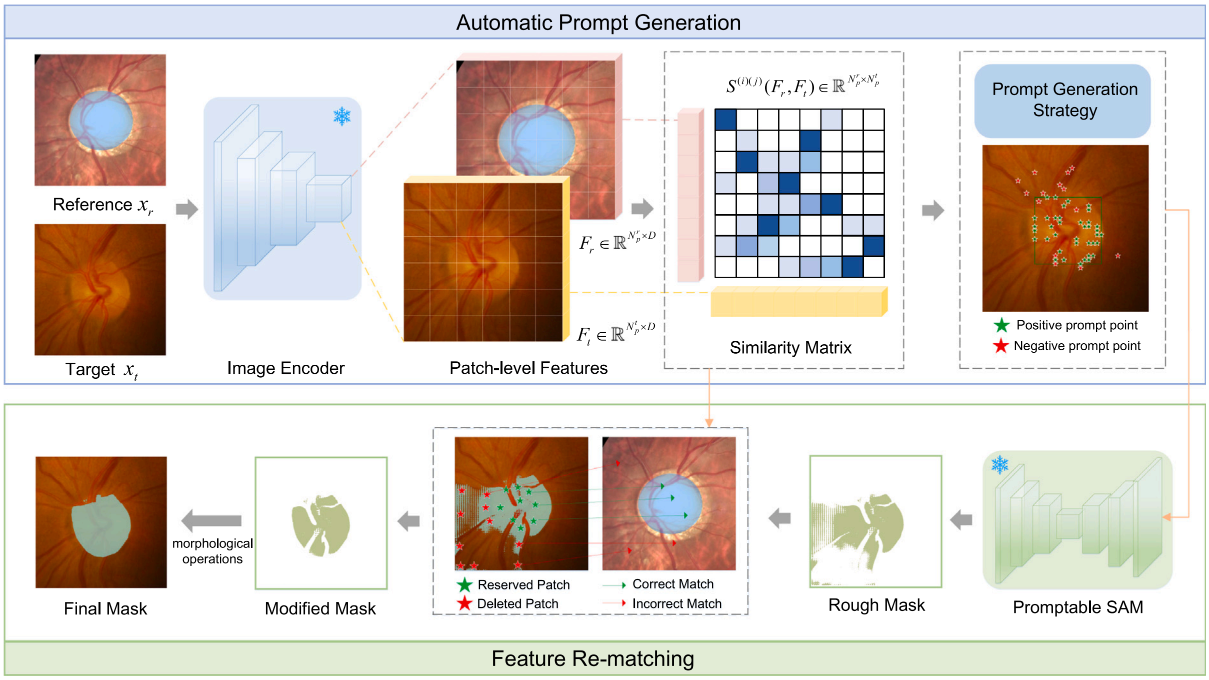

Fig. 1. The training-free overall framework of OSAM-Fundus requires only a fundus image annotation pair as reference. we utilize DINOv2 as an image encoder to extract patch-level features, and elaborate a prompt generation strategy to automatically obtain prompts on the target image based on the feature similarity matrix. Additionally, a feature re-matching module is introduced to correct the false-positive pixels potentially produced by SAM.

the same semantics in the target image. The overview of OSAM-Fundus is illustrated in Fig. 1. Our framework consists of two modules: Automatic Prompt Generation (APG) and Feature Re-matching (FR). First, we employ the general feature extractor DINOv2 to extract patch-level features of reference image x_r and target image x_t, computing the feature similarity matrix between all patches on x_r and x_t. Then we designed a prompt generation strategy based on the patch feature similarity matrix to automatically generate some accurate positive-negative matching points. These points serve as input to the prompt encoder in SAM to obtain a rough mask. Finally, we perform feature re-matching between the image patches corresponding to the reference mask and the rough mask to eliminate false-positive pixels. We extend to K-shot setting by independently applying our one-shot approach to each of the k images, resulting in k segmentation masks. Then, we assign weights to these masks based on image-level similarity and perform pixel-level voting fusion.

3.1. Automatic prompt generation

Our proposed automatic prompt generation module extracts patch-level features to form a similarity matrix, enabling the design of a prompt generation strategy to produce high-quality prompts for SAM.

3.1.1. Patch-wise feature similarity matrix construction

As shown in Fig. 1, given inputs reference image x_r and target image x_t, we apply DINOv2 as the image encoder Enc to extract patch-level features as:

$$F_r = \mathrm{Enc}(x_r), F_t = \mathrm{Enc}(x_t), \tag{1}$$

where $F_r \in \mathbb{R}^{N_p^r \times D}$ and $F_t \in \mathbb{R}^{N_p^t \times D}$, with N_p^r and N_p^t denote the number of patches of x_r and x_t respectively, and D denotes the embedding dimension of DINOv2. Then, the patch-wise similarity of the two features is computed for patch matching between images. We define the patch-wise similarity matrix $S \in \mathbb{R}^{N_p^r \times N_p^t}$ as:

$$S_{ij} = \| f_r^i - f_t^j \|, \tag{2}$$

where S_{ij} denotes the Euclidean distance between i-th patch feature f_r^i of F_r and j-th patch feature f_t^j of F_t. The similarity matrix can be applied either between different images to identify similar regions between them, or within the same image to filter specific regions within the image.

3.1.2. Prompt generation strategy

Although it is possible to directly select the most similar patches as prompts based on the similarity matrix S, this simple approach may lead to numerous outliers. Therefore, we design a high-quality prompt generation strategy based on the feature similarity matrix, aiming to guide SAM to segment the target region more precisely. The strategy consists of four processes, as shown in Fig. 2.

Forward matching. Given the reference mask m_r, first determine the set of foreground patch indices $P_r = \{i_k\}_{k=1}^{M}$, where M denotes the number of foreground patches. For each index i_k in the set P_r, the Euclidean distance between the patch feature $f_r^{i_k}$ and all the patch features of the target image f_t^j is computed to generate a forward similarity matrix $S^{\rightarrow}$. This can be represented as:

$$S_{ij}^{\rightarrow} = \| f_r^{i_k} - f_t^j \|, \text{ where } k = 1, 2, \ldots, M, j = 1, 2, \ldots, N_p^t. \tag{3}$$

Then, the set of target patch indices $P_t = \{j_k\}_{k=1}^{M}$ is obtained through nearest-neighbor matching.

Reverse matching. The low contrast between the foreground and background of the fundus image leads to outliers from vanilla forward matching, causing inaccuracies in segmentation results, as evident in Section 4.4. Inspired by the literature [60], we use the points in x_t obtained from forward matching to match the points in x_r, effectively removing the outliers. For each patch $j_k \in P_t$ of the target image, calculate the reverse similarity matrix $S^{\leftarrow}$,

$$S_{ij}^{\leftarrow} = \| f_r^i - f_t^{j_k} \|, \text{ where } i = 1, 2, \ldots, N_p^r, k = 1, 2, \ldots, M, \tag{4}$$

select the patches $P_r^{\leftarrow} = \{i_k{}^{\leftarrow}\}_{k=1}^{M}$ with the minimum distance as the reverse matching points in x_r, as shown in Fig. 2. Next, refine the forward matching set P_t by keeping the points that make the reverse match $P_r^{\leftarrow}$ in the foreground region, and obtain the corrected $P_t' =$

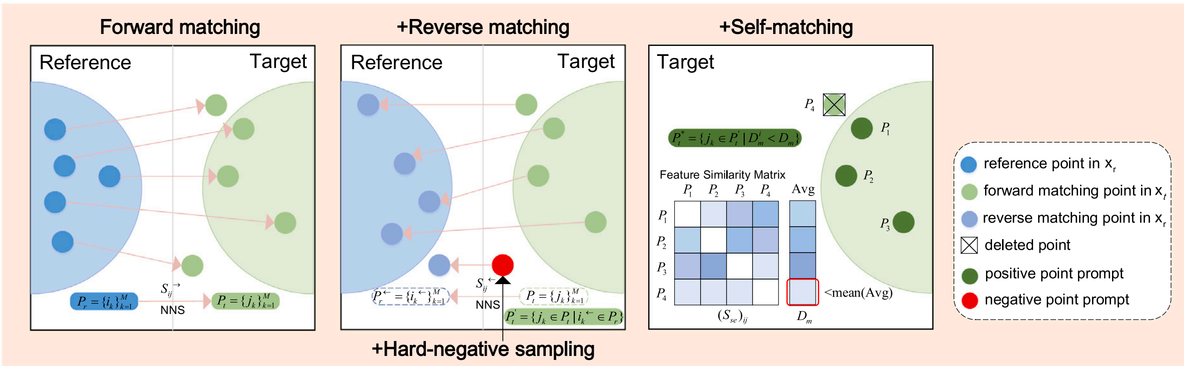

Fig. 2. Illustration of the four processes in the prompt generation strategy. These include forward matching, reverse matching, self-matching, and hard-negative sampling to gradually obtain an accurate prompt solution.

$\{j_k\}_{k=1}^{N} = \{j_k \in P_t \mid i_k^{\leftarrow} \in P_r\}$, where N is the number of retained patches.

Self-matching. In medical image analysis, the internal within the same anatomical region often exhibits consistent feature expressions. Based on this observation, we develop a self-matching mechanism to refine the P_t' by evaluating the similarity of matched patches in the target image. Specifically, we first calculate the feature similarity matrix S_{se} among the matched patches on x_t as:

$$(S_{se})_{ij} = \| f_r^{j_k} - f_t^{j_l} \|, \text{ where } i,j = 1,2,\ldots,N; i \neq j; k,l = 1,2,\ldots,N. \quad (5)$$

Then, for each patch $j_k \in P_t'$, we compute the average neighbor distance to other patches as $D_m^i = \frac{1}{N}\sum_{j\neq i}(S_{se})_{ij}$, and determine the overall average distance as $D_m = \frac{1}{N}\sum_i D_m^i$. Patches are then selected based on their average neighbor distance being less than the overall average distance, leading to the update of $P_t^* = \{j_k\}_k^L = \{j_k \in P_t' \mid D_m^i < D_m\}$, where L is the number of retained patches.

Hard-negative sampling. During the reverse matching process, we remove those points matching the background of the reference image from the outcomes of the forward matching P_t. In reality, these points are hard-negative samples that most closely resemble positive samples. To more effectively guide SAM to segment semantically ambiguous pixels, we intentionally select these points as negative prompt points to enhance the model's ability to understand complex image structures.

Finally, we add the bounding box of P_t^* as a box prompt. In practice, we found that adding the bounding box prompt leads to relatively precise OC and OD masks.

3.2. Feature re-matching

Following the acquisition of diverse prompts via APG, we employ the promptable SAM for segmentation. However, our evaluation and recent research indicate that directly applying promptable SAM to medical image segmentation fails to achieve satisfactory performance [30], as shown in Fig. 1. This limitation mainly arises from the significant domain gap between natural and medical images. Specifically, medical images have relatively ambiguous boundaries, leading to an increase in false-positive mask regions. Therefore, we design a feature re-matching module to enhance the precision of the initial mask.

Initially, for the patch set P_t^s corresponding to the initial mask of x_t, a forward matching process is conducted by analyzing the feature similarity matrix. During this process, segmentation outliers positioned in the background of x_t are difficult to match to the foreground of x_r, making it a relatively simple task to identify these outliers. Subsequently, these segmentation outliers are removed to refine the mask accuracy. Since SAM tends to segment all boundary regions related to the given prompts, it may segment the fundus image into different regions based on blood vessels. To obtain the final target mask, we expand and erode the mask through morphological closing operations to enhance the segmentation result.

3.3. Extension to K-shot

When the task is extended to the K-shot ($K > 1$) setting, multiple annotated reference images are available. Given a set of reference images $\{x_r^i\}_{i=1}^K$ along with their respective masks $\{m_r^i\}_{i=1}^K$ and a target image x_t, the APG method is employed to generate prompts for SAM. Subsequently, SAM segments the x_t to produce a collection of mask proposals $\{m_t^j\}_{j=1}^K$. Next, we crop the foreground regions of the target and reference images using masks and extract local features with an image encoder as:

$$\{F_t^j\} = \mathrm{Enc}(\{x_t \circ m_t^j\}), \{F_r^i\} = \mathrm{Enc}(\{x_r^i \circ m_r^i\}), \quad (6)$$

where $\circ$ denotes spatial-wise multiplication. After this, we calculate the similarity between $\{F_t^j\}$ and $\{F_r^i\}$, and assign a weight w_i to each reference image based on the similarity as:

$$w_i = \frac{S(F_t^j, F_r^i)}{\sum_{i=1}^K S(F_t^j, F_r^i)}, \quad (7)$$

where $S(F_t^j, F_r^i)$ denotes the similarity between $\{F_t^j\}$ and $\{F_r^i\}$, with $i = j$. Using these weights, we perform a weighted voting merge of the masks of all the reference images to generate a preliminary fusion mask as:

$$\hat{m} = \begin{cases} 1, & \text{if } \sum_{i=1}^K w_i \cdot m_r^i \geq \sum_{i=1}^K w_i \cdot (1 - m_r^i) \\ 0, & \text{otherwise} \end{cases} \quad (8)$$

On top of this, we extract global features from the original target image x_t and the reference image $\{x_r^i\}$, calculate the similarity between these global features, and select the reference image that is most similar to the target image. Finally, a feature re-matching operation between $\hat{m}$ and the dynamically selected reference image is performed to obtain the final refined mask.

4. Experiments

This section discusses the experimental details and results. We first present the datasets and evaluation metrics used in this work in Section 4.1. Secondly, the implementation details are provided in Section 4.2. Section 4.3 compares the performance of our approach on benchmark datasets with that of the baseline model and other state-of-the-art one-shot segmentation models. Section 4.4 displays the ablation studies.

4.1. Datasets and evaluation metrics

The optic disc and cup segmentation performance of OSAM-Fundus is evaluated on the public datasets REFUGE-train, REFUGE-val [17], Drishti-GS [15] and RIM-ONE-r3 [16], comprising retinal fundus images from 4 different clinical centers. The sample sizes are 400, 400,

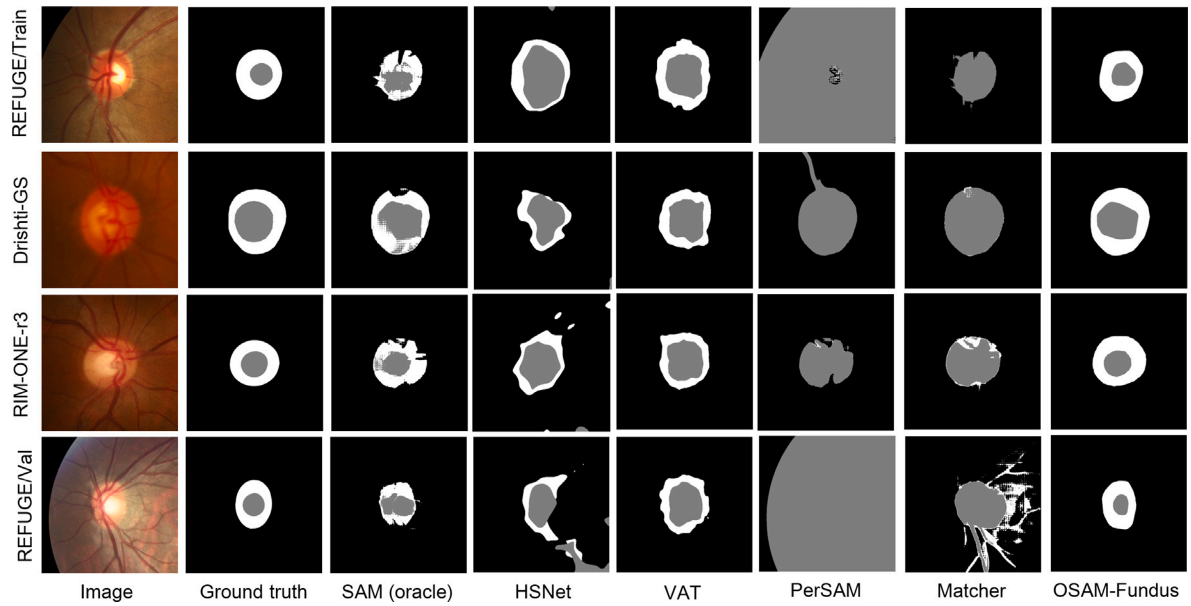

Fig. 3. Qualitative comparison of segmentation results among various methods in fundus optic disc and optic cup segmentation.

101, and 159, respectively. Varying acquisition equipment results in a degree of domain shift among datasets, providing a means to assess the segmentation accuracy within and outside the domain. During preprocessing, the images were uniformly cropped to an optic disc area of 800×800 pixels, subsequently resized to 560×560 to accommodate network input.

4.2. Implementation details

OSAM-Fundus adopts DINOv2 with ViT-G/14 [61] as a general -purpose image encoder. Thanks to unsupervised comparative learning on large-scale unlabeled datasets, DINOv2 demonstrates excellent feature representation capabilities and effectively promotes accurate patch matching between different images. Additionally, SAM with ViT-L serves as the segmenter of OSAM-Fundus. SAM exhibits compelling zero-shot segmentation performance following pre-training with 1B masks and 11M images. The combination of these powerful vision foundation models brings great potential for open-world image understanding. It is worth noting that no additional training is conducted on OSAM-Fundus in all experiments. This training-free feature gives OSAM-Fundus significant advantages in rapid deployment and adaptation to new tasks. The framework is implemented with Pytorch library and inferred on NVIDIA A40 GPU.

4.3. Comparison with previous methods

We employ the original SAM as our baseline and carefully select several state-of-the-art segmentation methods for comparison. These include the one-shot segmentation methods HSNet [43] and VAT [44], which require support images and their corresponding annotations for training. Additionally, we compare with training-free one-shot segmentation methods PerSAM [59] and Matcher [60], which require a reference image and corresponding annotation for guiding model inference. To further highlight the advantages of our method, we also compare it with fully-supervised segmentation methods, including the classical models U-Net [10] and Deeplabv3+ [62], as well as expert models BGA-Net [18] and AADG [23], which are specifically designed for fundus OD/OC segmentation and trained on 320 labeled images.

4.3.1. Experimental setting

We sample reference images from the training sets of the dataset separately and evaluate the segmentation metrics on their corresponding test sets. The division of the training and test sets follows the settings in previous literature [21]. To compare both in-domain and out-of-domain segmentation metrics, we present segmentation metrics on test sets from four different datasets, using the training set of the REFUGE-train as a reference gallery.

For the baseline method, we use an oracle mechanism that samples positive and negative prompt points directly from the ground truth mask. The number of samples corresponds to those generated by our automatic prompt generation, and the prompt type remains consistent with our method, using point and box composite prompt. To ensure fair comparisons, the support or reference images required for all comparison methods are consistent with our method.

4.3.2. Experiment results

Comparison with the baseline method. Despite being primarily trained on natural images, SAM demonstrates impressive capabilities when guided by oracle prompts. As shown in Fig. 3, SAM performs well on the OD segmentation task where a clear color contrast exists between the target and the background. However, SAM faces challenges on the OC segmentation task where the color closely resembles the background. As shown in Table 2, our OSAM-Fundus framework significantly outperforms the baseline on both the OD and OC segmentation tasks, achieving an average Dice improvement of 7.28%. This enhancement can be attributed to the accuracy of our automatic prompt generation method and feature re-matching module. Our proposed feature matching mechanism integrates medical prior knowledge, acknowledging that feature representations within the same anatomical structure tend to converge despite variations in appearance across images or domains. This convergence facilitates the accurate generation of prompts. Compared to the oracle SAM, which samples negative prompt points from the background, our hard negative sampling strategy acquires negative prompt points that are easily confused with foreground semantics, thereby providing better guidance for SAM segmentation. Additionally, the feature re-matching module can refine potential false-positive results generated by SAM, thereby further enhancing segmentation accuracy.

Table 2
Comparative experiments with other state-of-the-art segmentation methods on fundus OD and OC segmentation tasks, including the baseline method (SAM), one-shot segmentation methods (HSNet, VAT), and training-free one-shot segmentation methods (PerSAM, Matcher). (Dice% ↑).

Method	In-domain		Out-of-domain						Overall
	REFUGE/Train		Drishti-GS		RIM-ONE-r3		REFUGE/Val		
	OD	OC	OD	OC	OD	OC	OD	OC	
23'SAM (oracle)	90.99	68.79	90.39	72.90	69.97	41.88	85.25	65.06	73.15
21'HSNet 1-shot	81.87	38.20	79.13	62.78	66.29	38.78	65.29	28.78	57.64
22'VAT 1-shot	87.33	55.30	88.76	78.29	77.16	47.76	69.62	37.75	67.75
23'PerSAM 1-shot	26.27	8.42	66.94	43.05	39.35	23.65	14.85	4.31	28.36
24'Matcher 1-shot	85.92	44.25	91.23	58.18	69.16	35.49	62.95	42.87	61.26
OSAM-Fundus 1-shot	93.92	72.40	95.06	73.33	81.43	64.76	91.17	71.33	80.43
OSAM-Fundus 3-shot	94.08	77.53	95.16	76.41	82.00	68.15	91.51	73.76	82.33
OSAM-Fundus 5-shot	94.14	77.84	95.35	77.61	82.64	68.85	91.87	75.95	83.03
OSAM-Fundus 10-shot	**94.21**	**78.21**	**95.36**	**79.74**	**83.02**	**70.12**	**92.01**	**77.21**	**83.74**

Table 3
Comparative experiments with other fully-supervised segmentation methods, including classic models (U-net, DeepLabv3+) and expert models (BGA-Net, AADG). The results for the other methods were obtained using 320 fully labeled images for training, while our method requires no training and utilizes only 10 images. (Dice% ↑).

Method	In-domain		Out-of-domain						Overall
	REFUGE/Train		Drishti-GS		RIM-ONE-r3		REFUGE/Val		
	OD	OC	OD	OC	OD	OC	OD	OC	
15'U-Net	95.68	84.93	80.36	52.47	48.60	33.55	79.80	33.32	63.59
18'DeepLabv3+	95.10	83.52	84.13	70.27	81.50	34.04	88.24	64.93	75.22
21'BGA-Net	**96.18**	88.00	93.61	71.49	**86.67**	68.61	89.25	68.31	82.77
22'AADG	96.08	**88.47**	94.43	**82.57**	84.30	68.77	**94.70**	**86.14**	**86.93**
OSAM-Fundus	94.21	78.21	**95.36**	79.74	83.02	**70.12**	92.01	77.21	83.74

Comparisons with one-shot segmentation methods. In Table 2, we evaluate both in-domain and out-of-domain datasets, demonstrating OSAM's excellent segmentation accuracy and generalization ability. Our method surpasses training-free one-shot segmentation methods, exhibiting an average dice coefficient improvement of 52.7% over PerSAM and 19.17% over Matcher. Additionally, it outperforms one-shot segmentation methods that require training, with an average dice coefficient improvement of 22.79% over HSNet and 12.68% over VAT. The limitation of PerSAM is that it only provides one positive point and one negative point as a prompt, making it difficult to segment the OC with ambiguous edge semantics. As shown in Fig. 3, the OC and OD areas almost completely overlap. In addition, PerSAM is also prone to misjudgment in OD segmentation tasks. Particularly evident in the REFUGE dataset, the high contrast of fundus images against black backgrounds leads to misinterpretation of clear-edged eyeball outlines as segmentation results. Matcher expands the number of prompt points, improving the performance of the OD segmentation task on some datasets. However, the boundaries of OC and OD are difficult to distinguish due to similar anatomy as well as unclear and complex boundaries. This leads to the inapplicability of OD segmentation and the failure of OC segmentation, as shown in Fig. 3. Our SAM-Fundus enhances Matcher by offering a more precise prompting scheme derived from medical prior knowledge and hard negative sampling. Additionally, it effectively corrects false positive regions potentially generated by SAM. Notably, our method effectively utilizes the K-shot reference, with segmentation accuracy gradually improving as the number of prompted samples increases, as shown in Table 2.

Comparisons with fully-supervised segmentation methods. In Table 3, we compare OSAM-Fundus with fully supervised segmentation methods that were trained with 320 labeled images. The results show that OSAM-Fundus is able to achieve comparable or even better segmentation results than fully-supervised learning methods without training. While classical models like U-Net and DeepLabv3+ slightly outperform OSAM-Fundus on in-domain datasets, they have weaker generalization capabilities and significantly degrade their performance on out-of-domain datasets. Specifically, our method outperforms U-Net by 20.15% and DeepLabv3+ by 8.52% in average segmentation accuracy. Compared with the expert models, OSAM-Fundus outperforms BGA-Net by 0.97% in terms of average dice coefficient and achieves competitive performance with AADG, all without the need for additional training. In particular, OSAM-Fundus performs particularly well in the Out-of-domain dataset. In the Drishti-GS dataset, it surpasses all fully supervised methods in both OD and OC segmentation, and achieves superior OC segmentation performance compared to other methods in the RIM-ONE-r3 dataset. In summary, OSAM-Fundus outperforms the classical models on average and exceeds or approaches expert models. Considering that it requires no training, this performance is highly competitive, especially on the out-of-domain dataset where it demonstrates excellent generalization ability. Although its segmentation accuracy is slightly lower than fully-supervised methods in some specific scenarios, OSAM-Fundus significantly reduces the cost of data labeling and training, making it highly advantageous for practical applications.

4.4. Ablation analysis

We conduct ablation studies to investigate three key questions about OSAM-Fundus: (1) the contribution of individual components to model performance, (2) the influence of different prompt types on model performance, and (3) the effect of different backbones on model performance.

4.4.1. Contribution of each component

The OSAM-Fundus framework comprises two key modules: Automatic Prompt Generation (APG) and Feature Re-matching (FR). The prompt generation strategy of APG obtains a scheme containing positive and negative prompt points through four steps: forward matching, reverse matching, self-matching, and hard-negative sampling. Reverse matching relies on matching points on the target image obtained from forward matching, while self-matching further refines the matching results. Hard-negative sampling selects points deleted in reverse matching. The points generated by APG are then input into SAM to obtain initial masks, and feature re-matching refines them. Therefore, each

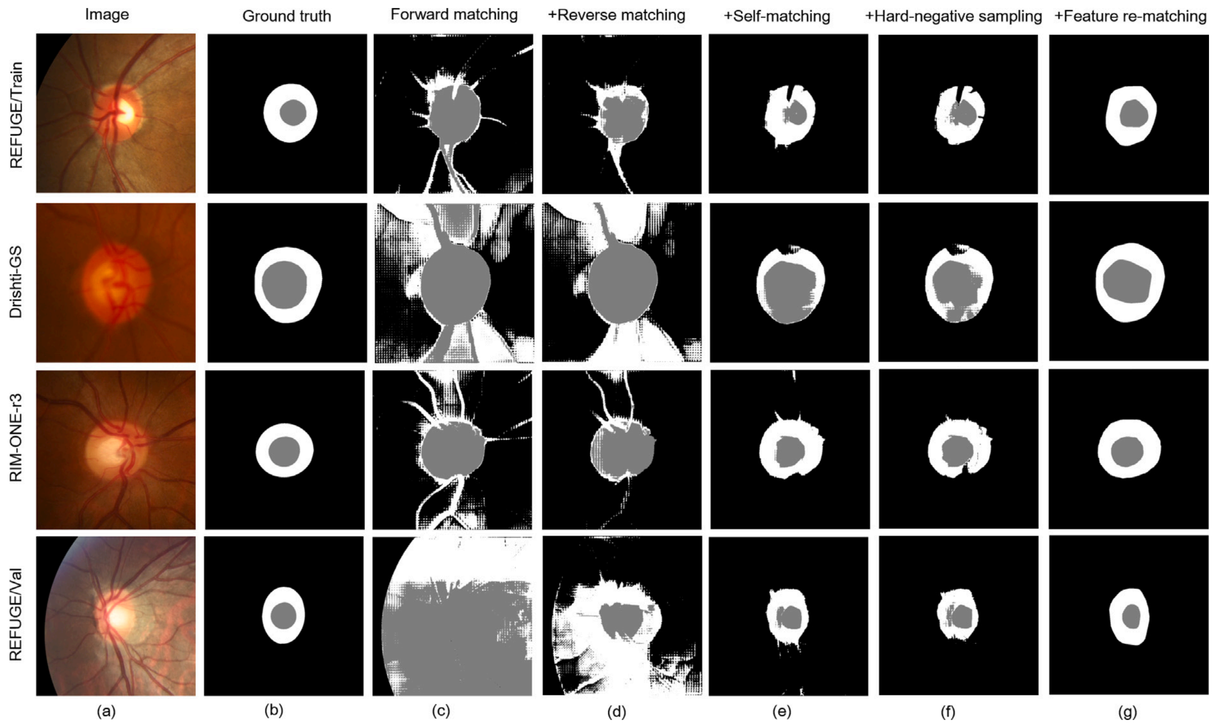

Fig. 4. Visualization of component contributions for ablation studies.

Table 4
Ablation study of component contributions to OD and OC segmentation in in-domain and out-of-domain settings (Dice% ↑).

Module	Component	In-domain		Out-of-domain						Overall
		REFUGE/Train		Drishti-GS		RIM-ONE-r3		REFUGE/Val		
		OD	OC	OD	OC	OD	OC	OD	OC	
APG	Forward matching	38.14	31.33	47.20	48.62	33.53	32.62	15.77	19.25	33.31
	+Reverse matching	75.94	52.26	62.11	57.75	50.82	50.60	46.27	50.40	55.77
	+Self-matching	91.69	69.66	90.32	69.12	74.84	59.50	85.48	63.67	75.54
	+Hard-negative sampling	90.47	70.40	88.91	70.22	81.10	62.07	85.73	67.52	77.06
FR	+Feature re-matching	**93.92**	**72.40**	**95.06**	**73.33**	**81.43**	**64.76**	**91.17**	**71.33**	**80.43**

component is progressive, so we observe changes in model performance by gradually introducing these components.

As shown in Table 4, each component contributes to an improvement in segmentation accuracy, with corresponding visual results depicted in Fig. 4. Specifically, the APG module gradually improves the accuracy of feature matching, ensuring the generation of precise prompts. Forward matching alone is prone to generate numerous matching outliers, resulting in more false-positive regions, as shown in Fig. 4(c). Reverse matching improves the accuracy of feature matching and effectively reduces the false-positive segmentation results, but it challenges remain in distinguishing the OD and OC boundaries, as seen in Fig. 4(d). We designed self-matching (Fig. 4(e)) based on the similarity of features within the same anatomical structure, which is able to filter out the points with inconsistent features and further reduce false matches. For semantically ambiguous edge regions in medical images, hard-negative sampling (Fig. 4(f)) selects easily confused background points as negative prompts, effectively guiding SAM's segmentation of the foreground and background. The prompting scheme generated by APG is fed into SAM to produce an initial rough mask. However, the limitation of SAM in dealing with fuzzy edges become evident, as shown in Fig. 4(f). To address this, the FR module refines the rough mask by removing false positive pixels to obtain a more accurate segmentation output, as shown in Fig. 4(g).

4.4.2. Analysis of SAM prompt types

SAM currently supports two prompt types, point prompt and box prompt. In Table 5, the effect of a single prompt type remains consistent for the more ambiguous optic cup. However, for optic disc with clearer contours, the point prompt demonstrates significantly higher accuracy compared to the box prompt. This discrepancy may arise from the susceptibility of the box prompt to imperfect matching results. Specifically, while a small number of outliers minimally impact the performance of the segmentation model, the box prompts derived from these outliers can provide negative guidance. In contrast, the composite prompt type, which integrates point and box prompts, exhibits superior model performance. The composite prompt provides more structured information and context, enabling the model to better understand the location and boundaries of the target region. Especially in the processing of medical images with fuzzy boundaries and complex structures, this prompt type proves more effective.

Table 5
Ablation study of different prompt types for SAM in in-domain and out-of-domain settings (Dice% ↑).

Prompt type	In-domain		Out-of-domain					
	REFUGE/Train		Drishti-GS		RIM-ONE-r3		REFUGE/Val	
	OD	OC	OD	OC	OD	OC	OD	OC
point	89.70	49.54	85.00	59.13	67.78	42.33	81.55	48.40
box	80.69	57.40	81.10	48.63	77.07	40.98	64.98	52.76
point+box	**90.47**	**70.40**	**89.55**	**69.16**	**81.10**	**62.07**	**85.73**	**67.52**

Table 6
Ablation study of different backbones of SAM in in-domain and out-of-domain settings (Dice% ↑).

Backbone	In-domain		Out-of-domain					
	REFUGE/Train		Drishti-GS		RIM-ONE-r3		REFUGE/Val	
	OD	OC	OD	OC	OD	OC	OD	OC
VIT-B	67.14	56.73	72.43	61.87	59.36	53.50	41.93	49.13
VIT-L	**90.47**	**70.40**	89.56	69.16	81.10	**62.07**	**85.73**	**67.52**
VIT-H	88.94	66.92	**91.65**	**69.86**	**81.88**	60.81	79.94	64.28

4.4.3. Analysis of different segmentation backbone

SAM provides three different image encoders: ViT-B, ViT-L, and ViT-H. These encoders vary in parameter counts, with ViT-B having 91M, ViT-L having 308M, and ViT-H having 636M parameters. In general, models with larger parameters tend to achieve better segmentation results due to their stronger feature representation capabilities. However, in the segmentation task of the fundus optic disc and optic cup, we observed that ViT-L outperformed ViT-H in most cases, as depicted in Table 6. This outcome may be due to the high-resolution feature representation of ViT-H, which enhances the model's sensitivity to edges of non-target regions, consequently resulting in more fragmented segmentation masks.

5. Discussion

SAM has demonstrated excellent performance in various natural image segmentation tasks. Recently, efforts have been made to extend SAM to medical image segmentation. However, as an interactive model, SAM requires more manual interventions and specific medical expertise during medical image segmentation, consequently diminishing its operational efficiency. Moreover, the ambiguous and intricate boundaries in medical images often lead to the generation of masks that include non-target regions, even with more accurate prompts as inputs. Fine-tuning SAM on various medical image segmentation tasks is currently one of the most common solutions. However, such methods still rely on a large number of annotations for training, and the fine-tuning process itself reduces the generalization ability of the model.

To address these issues, we propose OSAM-Fundus, which aims to achieve OD and OC segmentation of all fundus images from a single reference annotation without any additional human intervention or training. Our proposed APG module generates prompt based on the reference annotations to guide the segmentation process, which significantly reduces the dependence on expert knowledge and greatly simplifies the task of interactive medical image segmentation. As shown in Table 4, adding reverse matching to forward matching can initially correct the matching results. Self-matching, enhanced with medical prior knowledge, is the key factor in achieving accurate segmentation. This step reduces the errors that may arise from transferring knowledge from natural images to the fundus segmentation task. To further enhance the semantic understanding of the boundary region, we chose confusing background points as negative prompt points to further improve the segmentation accuracy. Furthermore, to address the common fuzzy boundary problem in medical images, we introduce a feature rematching module, which effectively improves segmentation accuracy by eliminating potential false-positive results.

Our experimental results confirm the superiority of OSAM-Fundus. We conduct experiments on four in-domain and out-of-domain datasets, each collected by different organizations and varying in scanner vendors, patient populations, field of view, and image appearance. Despite these substantial differences, OSAM-Fundus consistently achieves superior performance. Compared to fully-supervised learning methods, OSAM-Fundus performs outstandingly well without requiring large amounts of annotated data, and the model generates high-quality segmentation masks even under changing conditions. In terms of overall performance, OSAM-Fundus not only outperforms classical models, but also approaches or exceeds expert models. Specifically, our method surpasses U-Net by 20.15%, Deeplabv3+ by 8.52%, and the expert model BGA-Net by 0.97% in average dice coefficient. Compared to one-shot segmentation methods that require training, our model is not only more accurate but also reduces training time. Specifically, OSAM-Fundus improves the average dice coefficient by 22.79% over HSNet and 12.68% over VAT. Furthermore, compared to training-free one-shot segmentation methods, our method effectively transfers knowledge from the natural image segmentation domain to the medical image segmentation, with an average Dice coefficient improvement of 52.07% over PerSAM and 19.17% over Matcher. These results demonstrate that OSAM-Fundus exhibits robust performance under diverse conditions while significantly reducing both training and annotation costs.

OSAM-Fundus comprises 1442.26M parameters and has 3130.20G FLOPs, with an input resolution of 560×560 for feature extraction and 1024×1024 for segmentation. During inference, it requires 9236 MB of GPU memory and 4253 MB of CPU memory, which is a reasonable requirement for the commonly used computing hardware in clinical research. Additionally, the model's average inference time per image is 1.59 s, enabling efficient processing. As a companion diagnostic tool, OSAM-Fundus can be integrated with the hospital imaging system to automatically analyze and segment the optic disc and cup regions in fundus images and calculate their cup-to-disc ratios to screen for possible glaucoma cases. Ophthalmologists can use the segmentation and quantitative results provided by the system, along with other clinical information, to make faster and more accurate diagnostic decisions.

In exploring the effects of different prompt types, we discovered that the composite prompt (combining points and boxes) generally outperforms the single prompt. This finding provides valuable insights for other medical image segmentation tasks. Furthermore, our ablation experiments reveal a non-intuitive finding: the best performance in medical image segmentation tasks is not always achieved by models with more parameters. In fact, due to the complexity of medical image edges, highly sensitive models may incorrectly segment non-target regions.

Finally, the limitations of SAM-Fundus need to be discussed. Although our proposed feature re-matching module can effectively eliminate the erroneous segmentation masks of SAM, it may generate outliers that lead to the removal of true-positive pixels. While these erroneously deleted pixels can be recovered by the closure operation in morphology when segmenting structures with closed shapes, it poses challenges for radial structure segmentation like blood vessel segmentation. As part of future work, further efforts are needed to explore more suitable false-positive mask elimination methods for radial structure segmentation. Additionally, we acknowledge the limitations of SAM in scenarios lacking clear color contrast between foreground and background. For example, as shown in Fig. 5(b), the small luminance difference between the OD's edge region and the surrounding background causes difficulties in handling such fuzzy boundaries, reducing the segmentation accuracy in the OD region. Similarly, the brightness and structural similarities between the OC and OD make it challenging to accurately distinguish their boundaries, resulting in segmentation errors in the OC region. To this end, we propose potential directions for future enhancements and improvements.

5.1. Building large-scale medical datasets

The success of SAM in natural scenes is largely supported by numerous benchmark datasets. However, the segmentation annotation of medical datasets often requires domain-specific expertise and has relatively complex boundaries and finer anatomical structures, making building large-scale medical datasets a labor-intensive and time-consuming process. While the segmentation results generated by SAM are not always perfect, its interactive nature allows for targeted modification of details without the need to label targets from scratch. This development significantly reduces the annotation costs associated with

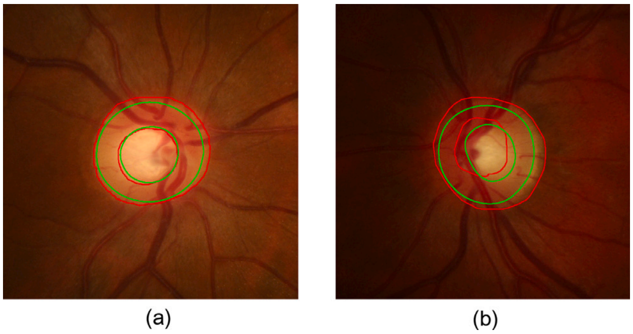

Fig. 5. Segmentation Results for Fundus Images. (a) Successful case. (b) Challenging case. The red contours represent the prediction boundaries, and the green contours are ground truths. The average Dice coefficient is 91.68% for (a) and 76.36% for (b).

creating large-scale datasets. Recently, several studies have published large-scale medical datasets aimed to facilitate the development of medical image analysis by collecting and standardizing public datasets. Based on these datasets, pre-trained models in the medical image field will continue to be enhanced to improve semantic understanding in complex segmentation scenes, such as those with extreme blurring or low contrast. Our approach can work with these pre-trained models to achieve even better performance.

5.2. Fine-tuning based on weakly supervised learning

Currently, many studies focus on either fully fine-tuning SAM or tuning some of its parameters using various parameter-efficient fine-tuning techniques. While this approach has been proven effective, it still requires a large amount of labeled data for training, which does not reduce the labor cost of data labeling. Although SAM achieved sub-optimal results in some low-contrast scenarios, the segmentation masks it generates can serve as pseudo-labels to provide weakly supervised information. The weakly supervised learning strategy allows the model to learn from imperfect labels while progressively optimizing its segmentation capabilities as the data size increases and the model continues to iterate. Therefore, we believe this is a promising solution.

In the future, we will explore the application of OSAM-Fundus on a wider range of medical image datasets to verify its applicability and effectiveness in different clinical situations.

6. Conclusion

In this study, we present OSAM-Fundus, a novel training-free fundus image segmentation framework that requires only a one-reference image and its mask for effective segmentation of the optic disc and optic cup. Specifically, we design an automatic prompt segmentation method that does not require any additional manual operations. Our proposed method achieves cross-image and cross-domain semantic matching by extracting generic visual features and performing patch-level bi-directional feature matching. To obtain a more accurate segmentation prompt guidance scheme, we incorporate medical image prior knowledge and hard-negative sampling. This approach effectively realizes knowledge transfer from natural to medical images for general-purpose visual models, and significantly reduces the dependence of interactive segmentation models on manual operations and medical expertise. To address the limitations of SAM in segmenting fuzzy boundaries, we refine the segmentation results by feature rematching to get the output of the same semantic objects in the target image.

An important challenge in the fundus segmentation task is domain shift, denoting the substantial decline in the model's performance when faced with unseen domains diverging from the distribution of the training data. Our method relies on one-shot reference images to achieve an average dice coefficient of 80.43% on four fundus datasets, which is the state-of-the-art for one-shot segmentation without training. The method shows an 7.28% improvement over SAM when using prompts generated from ground truth. Compared to other one-shot segmentation methods that require training, OSAM-Fundus not only delivers superior accuracy but also significantly reduces training time costs. Additionally, unlike fully supervised learning methods that depend on extensive labeled data, OSAM-Fundus achieves excellent segmentation results without such dependency and maintains robust performance across varying conditions. These results shows that OSAM-Fundus is an efficient and robust method for fundus image segmentation.

Additionally, OSAM-Fundus has a wide range of clinical applications. It can be integrated as an assistant segmentation tool within hospital imaging systems to automatically process fundus images in the database, generating segmentation results of optic disc and cup along with quantitative analyses, thereby providing critical references for doctors' diagnoses.

In summary, our study highlights the potential of one-shot prompting techniques for precise fundus image analysis, offering opportunities to minimize the expenses associated with data labeling and training. Future work will focus on further optimize OSAM-Fundus to cope with its limitations in segmenting radial structures and low-contrast scenes. To this end, we propose two potential research directions. Firstly, to construct large-scale medical datasets to promote the continuous development of pre-trained models and improve their semantic understanding of complex medical images. Secondly, to explore the fine-tuning methods based on weakly-supervised learning, which can reduce the dependence on labeled data and enhance model generalization. These efforts are expected to further improve the applicability and effectiveness of OSAM-Fundus across various clinical settings.

CRediT authorship contribution statement

Rui Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Zhouwang Yang:** Writing – review & editing, Writing – original draft, Funding acquisition, Conceptualization. **Yanzhi Song:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] T. Li, W. Bo, C. Hu, H. Kang, H. Liu, K. Wang, H. Fu, Applications of deep learning in fundus images: A review, Med. Image Anal. 69 (2021) 101971.

[2] Y. Bazi, M.M. Al Rahhal, H. Elgibreen, M. Zuair, Vision transformers for segmentation of disc and cup in retinal fundus images, Biomed. Signal Process. Control 91 (2024) 105915.

[3] G.D. Joshi, J. Sivaswamy, S. Krishnadas, Optic disk and cup segmentation from monocular color retinal images for glaucoma assessment, IEEE Trans. Med. Imaging 30 (6) (2011) 1192–1205.

[4] M.C.V.S. Mary, E.B. Rajsingh, J.K.K. Jacob, D. Anandhi, U. Amato, S.E. Selvan, An empirical study on optic disc segmentation using an active contour model, Biomed. Signal Process. Control 18 (2015) 19–29.

[5] V.V.N.S. Kumar, G.H. Reddy, M. GiriPrasad, A novel glaucoma detection model using unet++-based segmentation and ResNet with GRU-based optimized deep learning, Biomed. Signal Process. Control 86 (2023) 105069.

[6] J. Zilly, J.M. Buhmann, D. Mahapatra, Glaucoma detection using entropy sampling and ensemble learning for automatic optic cup and disc segmentation, Comput. Med. Imaging Graph. 55 (2017) 28–41.

[7] D. Mohan, J.H. Kumar, C.S. Seelamantula, High-performance optic disc segmentation using convolutional neural networks, in: 2018 25th IEEE International Conference on Image Processing, ICIP, IEEE, 2018, pp. 4038–4042.

[8] J. Long, E. Shelhamer, T. Darrell, Fully convolutional networks for semantic segmentation, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015, pp. 3431–3440.

[9] S.M. Shankaranarayana, K. Ram, K. Mitra, M. Sivaprakasam, Fully convolutional networks for monocular retinal depth estimation and optic disc-cup segmentation, IEEE J. Biomed. Health Inform. 23 (4) (2019) 1417–1426.

[10] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional networks for biomedical image segmentation, in: Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015, Proceedings, Part III 18, Springer, 2015, pp. 234–241.

[11] H. Fu, J. Cheng, Y. Xu, D.W.K. Wong, J. Liu, X. Cao, Joint optic disc and cup segmentation based on multi-label deep network and polar transformation, IEEE Trans. Med. Imaging 37 (7) (2018) 1597–1605.

[12] X. Yuan, L. Zhou, S. Yu, M. Li, X. Wang, X. Zheng, A multi-scale convolutional neural network with context for joint segmentation of optic disc and cup, Artif. Intell. Med. 113 (2021) 102035.

[13] Z. Wei, P. Yuhan, J. Jianhang, Y. Jikun, B. Weiqi, Y. Yugen, W. Wenle, RMSDSC-net: a robust multiscale feature extraction with depthwise separable convolution network for optic disc and cup segmentation, Int. J. Intell. Syst. 37 (12) (2022) 11482–11505.

[14] Y. Yi, Y. Jiang, B. Zhou, N. Zhang, J. Dai, X. Huang, Q. Zeng, W. Zhou, C2FTFNet: Coarse-to-fine transformer network for joint optic disc and cup segmentation, Comput. Biol. Med. 164 (2023) 107215.

[15] J. Sivaswamy, S. Krishnadas, A. Chakravarty, G. Joshi, A.S. Tabish, et al., A comprehensive retinal image dataset for the assessment of glaucoma from the optic nerve head analysis, JSM Biomed. Imaging Data Pap. 2 (1) (2015) 1004.

[16] F. Fumero, S. Alayón, J.L. Sanchez, J. Sigut, M. Gonzalez-Hernandez, RIM-ONE: An open retinal image database for optic nerve evaluation, in: 2011 24th International Symposium on Computer-Based Medical Systems, CBMS, IEEE, 2011, pp. 1–6.

[17] J.I. Orlando, H. Fu, J.B. Breda, K. Van Keer, D.R. Bathula, A. Diaz-Pinto, R. Fang, P.-A. Heng, J. Kim, J. Lee, et al., Refuge challenge: A unified framework for evaluating automated methods for glaucoma assessment from fundus photographs, Med. Image Anal. 59 (2020) 101570.

[18] L. Luo, D. Xue, F. Pan, X. Feng, Joint optic disc and optic cup segmentation based on boundary prior and adversarial learning, Int. J. Comput. Assist. Radiol. Surg. 16 (6) (2021) 905–914.

[19] S. Wang, L. Yu, X. Yang, C.-W. Fu, P.-A. Heng, Patch-based output space adversarial learning for joint optic disc and cup segmentation, IEEE Trans. Med. Imaging 38 (11) (2019) 2485–2495.

[20] W. Zhou, J. Ji, W. Cui, Y. Wang, Y. Yi, Unsupervised domain adaptation fundus image segmentation via multi-scale adaptive adversarial learning, IEEE J. Biomed. Health Inf. (2023).

[21] S. Wang, L. Yu, K. Li, X. Yang, C.-W. Fu, P.-A. Heng, Dofe: Domain-oriented feature embedding for generalizable fundus image segmentation on unseen datasets, IEEE Trans. Med. Imaging 39 (12) (2020) 4237–4248.

[22] Q. Liu, C. Chen, J. Qin, Q. Dou, P.-A. Heng, Feddg: Federated domain generalization on medical image segmentation via episodic learning in continuous frequency space, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 1013–1023.

[23] J. Lyu, Y. Zhang, Y. Huang, L. Lin, P. Cheng, X. Tang, Aadg: automatic augmentation for domain generalization on retinal image segmentation, IEEE Trans. Med. Imaging 41 (12) (2022) 3699–3711.

[24] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A.C. Berg, W.-Y. Lo, et al., Segment anything, 2023, arXiv preprint arXiv:2304.02643.

[25] G. Ning, H. Liang, Z. Jiang, H. Zhang, H. Liao, The potential of'segment anything'(SAM) for universal intelligent ultrasound image guidance, BioSci. Trends (2023).

[26] S. Ren, F. Luzi, S. Lahrichi, K. Kassaw, L.M. Collins, K. Bradbury, J.M. Malof, Segment anything, from space? in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2024, pp. 8355–8365.

[27] L.P. Osco, Q. Wu, E.L. de Lemos, W.N. Gonçalves, A.P.M. Ramos, J. Li, J.M. Junior, The segment anything model (sam) for remote sensing applications: From zero to one shot, Int. J. Appl. Earth Obs. Geoinf. 124 (2023) 103540.

[28] L. Yao, H. Zuo, G. Zheng, C. Fu, J. Pan, Sam-da: Uav tracks anything at night with sam-powered domain adaptation, 2023, arXiv preprint arXiv:2307.01024.

[29] S. He, R. Bao, J. Li, J. Stout, A. Bjornerud, P.E. Grant, Y. Ou, Computer-vision benchmark segment-anything model (sam) in medical images: Accuracy in 12 datasets, 2023, arXiv preprint arXiv:2304.09324 3.

[30] M.A. Mazurowski, H. Dong, H. Gu, J. Yang, N. Konz, Y. Zhang, Segment anything model for medical image analysis: an experimental study, Med. Image Anal. 89 (2023) 102918.

[31] Y. Huang, X. Yang, L. Liu, H. Zhou, A. Chang, X. Zhou, R. Chen, J. Yu, J. Chen, C. Chen, et al., Segment anything model for medical images? Med. Image Anal. 92 (2024) 103061.

[32] J. Ma, Y. He, F. Li, L. Han, C. You, B. Wang, Segment anything in medical images, Nature Commun. 15 (1) (2024) 654.

[33] T. Ren, S. Liu, A. Zeng, J. Lin, K. Li, H. Cao, J. Chen, X. Huang, Y. Chen, F. Yan, et al., Grounded sam: Assembling open-world models for diverse visual tasks, 2024, arXiv preprint arXiv:2401.14159.

[34] S. Liu, Z. Zeng, T. Ren, F. Li, H. Zhang, J. Yang, C. Li, J. Yang, H. Su, J. Zhu, et al., Grounding dino: Marrying dino with grounded pre-training for open-set object detection, 2023, arXiv preprint arXiv:2303.05499.

[35] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, B. Ommer, High-resolution image synthesis with latent diffusion models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 10684–10695.

[36] J. Li, D. Li, C. Xiong, S. Hoi, Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation, in: International Conference on Machine Learning, PMLR, 2022, pp. 12888–12900.

[37] A. Shaban, S. Bansal, Z. Liu, I. Essa, B. Boots, One-shot learning for semantic segmentation, 2017, arXiv preprint arXiv:1709.03410.

[38] J. Snell, K. Swersky, R. Zemel, Prototypical networks for few-shot learning, Adv. Neural Inf. Process. Syst. 30 (2017).

[39] X. Zhang, Y. Wei, Y. Yang, T.S. Huang, Sg-one: Similarity guidance network for one-shot semantic segmentation, IEEE Trans. Cybern. 50 (9) (2020) 3855–3865.

[40] G. Li, V. Jampani, L. Sevilla-Lara, D. Sun, J. Kim, J. Kim, Adaptive prototype learning and allocation for few-shot segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 8334–8343.

[41] J. Liu, Y. Bao, G.-S. Xie, H. Xiong, J.-J. Sonke, E. Gavves, Dynamic prototype convolution network for few-shot semantic segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 11553–11562.

[42] Y. Liu, N. Liu, Q. Cao, X. Yao, J. Han, L. Shao, Learning non-target knowledge for few-shot semantic segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 11573–11582.

[43] J. Min, D. Kang, M. Cho, Hypercorrelation squeeze for few-shot segmentation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 6941–6952.

[44] S. Hong, S. Cho, J. Nam, S. Lin, S. Kim, Cost aggregation with 4d convolutional swin transformer for few-shot segmentation, in: European Conference on Computer Vision, Springer, 2022, pp. 108–126.

[45] K. Nguyen, S. Todorovic, Feature weighting and boosting for few-shot segmentation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 622–631.

[46] X. Li, T. Wei, Y.P. Chen, Y.-W. Tai, C.-K. Tang, Fss-1000: A 1000-class dataset for few-shot segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 2869–2878.

[47] A. Zhao, G. Balakrishnan, F. Durand, J.V. Guttag, A.V. Dalca, Data augmentation using learned transformations for one-shot medical image segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 8543–8553.

[48] A.G. Roy, S. Siddiqui, S. Pölsterl, N. Navab, C. Wachinger, 'Squeeze & excite' guided few-shot segmentation of volumetric images, Med. Image Anal. 59 (2020) 101587.

[49] R. Feng, X. Zheng, T. Gao, J. Chen, W. Wang, D.Z. Chen, J. Wu, Interactive few-shot learning: Limited supervision, better medical image segmentation, IEEE Trans. Med. Imaging 40 (10) (2021) 2575–2588.

[50] Q. Yu, K. Dang, N. Tajbakhsh, D. Terzopoulos, X. Ding, A location-sensitive local prototype network for few-shot medical image segmentation, in: 2021 IEEE 18th International Symposium on Biomedical Imaging, ISBI, IEEE, 2021, pp. 262–266.

[51] H. Tang, X. Liu, S. Sun, X. Yan, X. Xie, Recurrent mask refinement for few-shot medical image segmentation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 3918–3928.

[52] B. Min, H. Ross, E. Sulem, A.P.B. Veyseh, T.H. Nguyen, O. Sainz, E. Agirre, I. Heintz, D. Roth, Recent advances in natural language processing via large pre-trained language models: A survey, ACM Comput. Surv. 56 (2) (2023) 1–40.

[53] A. Radford, J.W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al., Learning transferable visual models from natural language supervision, in: International Conference on Machine Learning, PMLR, 2021, pp. 8748–8763.

[54] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby, et al., Dinov2: Learning robust visual features without supervision, 2023, arXiv preprint arXiv:2304.07193.

[55] C. Wang, D. Li, S. Wang, C. Zhang, Y. Wang, Y. Liu, G. Yang, SAM med: A medical image annotation framework based on large vision model, 2023, arXiv preprint arXiv:2307.05617 3.

[56] S. Na, Y. Guo, F. Jiang, H. Ma, J. Huang, Segment any cell: A SAM-based auto-prompting fine-tuning framework for nuclei segmentation, 2024, arXiv preprint arXiv:2401.13220.

[57] S. Cao, Q. Wu, L. Ma, TongueSAM: An universal tongue segmentation model based on SAM with zero-shot, in: 2023 IEEE International Conference on Bioinformatics and Biomedicine, BIBM, IEEE, 2023, pp. 4520–4526.

[58] C. Li, X. Han, C. Yao, X. Ban, MatSAM: Efficient materials microstructure extraction via visual large model, 2024, arXiv preprint arXiv:2401.05638.

[59] R. Zhang, Z. Jiang, Z. Guo, S. Yan, J. Pan, H. Dong, P. Gao, H. Li, Personalize segment anything model with one shot, 2023, arXiv preprint arXiv:2305.03048.

[60] Y. Liu, M. Zhu, H. Li, H. Chen, X. Wang, C. Shen, Matcher: Segment anything with one shot using all-purpose feature matching, 2023, arXiv preprint arXiv:2305.13310.

[61] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al., An image is worth 16x16 words: Transformers for image recognition at scale, 2020, arXiv preprint arXiv:2010.11929.

[62] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, H. Adam, Encoder-decoder with atrous separable convolution for semantic image segmentation, in: Proceedings of the European Conference on Computer Vision, ECCV, 2018, pp. 801–818.